

EE2213 Introduction to Artificial Intelligence

Lecture 5: Optimization I

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OVERVIEW OF COURSE CONTENTS

- **Introduction (Shaojing)**

- What is AI
- Applications of AI
- AI agent

Path optimization

- **Search (Shaojing)**

- Uninformed search algorithms: breadth-first, depth-first, uniform-cost(Dijkstra's algorithm)
- Informed search algorithms: greedy best-first, A*
- Applications

- **Optimization (Shaojing)**

- Linear programming
- Convex problems
- Applications

Mathematical
optimization

- **Machine learning (Wang Si)**

- Supervised and unsupervised learning: regression, classification, clustering
- Neural networks and deep learning
- Applications

- **Knowledge representation (Wang Si)**

- Knowledge Representation and Reasoning
- Propositional Logic
- Applications

- **Ethical considerations (Shaojing)**

- Bias in AI
- Privacy concerns
- Societal impact

Agenda

- Background of linear programs
- Recap of linear algebra
- Solving linear programs (LP)
 - The graphical method
 - Different cases of LP outcomes
 - The Simplex algorithm

Background

- **Optimization**

Choosing the best option from a set of options

- **Constrained optimization**

An optimization problem is constrained if solutions must satisfy some hard constraints on the values of the variables.

$$\underset{x}{\text{minimize}} \quad f(x)$$

$$\text{subject to } x \in X$$

The set X is called the feasible set.

➤ **Example:** minimize x^2 , s.t. $x \in [-1, 1]$

Background

- **Constraint types**

Generally, constraints are formulated using two types

1. Equality constraints: $h(x) = 0$
2. Inequality constraints: $g(x) \leq 0$

Any optimization problem can be written as

$$\underset{x}{\text{minimize}} \quad f(x)$$

$$\begin{aligned} \text{subject to} \quad & h_i(x) = 0 \text{ for all } i \text{ in } \{1, \dots, l\} \\ & g_j(x) \leq 0 \text{ for all } j \text{ in } \{1, \dots, m\} \end{aligned}$$

Linear programming problem

A special category of **constrained optimization** problem, where the constraints must be **linear inequalities**.

"Programming" originally meant **"resource planning"** in military context.

Now, "programming" simply means solving a mathematical problem.

Linear program example:

You want to plan your meals for the week to **minimize total cost** while meeting your nutritional needs for **calories** and **protein**.

Food	Cost (\$)	Calories	Proteins (g)
Chicken	3	250	30
Rice	1	200	4
Vegetables	2	50	2

Decision variables:

- x_1 : number of servings of chicken;
- x_2 : number of servings of rice
- x_3 : number of servings of vegetables

Objective:

$$\text{Minimize cost} = 3x_1 + 1x_2 + 2x_3$$

Constraints:

$$\text{Calories} \geq 2000: \quad 250x_1 + 200x_2 + 50x_3 \geq 2000$$

$$\text{Protein} \geq 50 \text{ grams:} \quad 30x_1 + 4x_2 + 2x_3 \geq 50$$

$$\text{Servings must be non-negative:} \quad x_1, x_2, x_3 \geq 0$$



Why linear programming is important?

- A **lot of problems** can be formulated as linear programs, and
- There exist **efficient methods** to solve them
- or at least give **good approximations**.

The transportation problem

There are **I factories** (index by i), each producing iPhone with a production capacity s_i (units)

There are **J markets** (indexed by j , e.g, the Singapore market, the US market, etc) demand iPhones, each with a demand d_j (units)

The cost of transporting one unit of iPhone from factory i to market j is c_{ij}

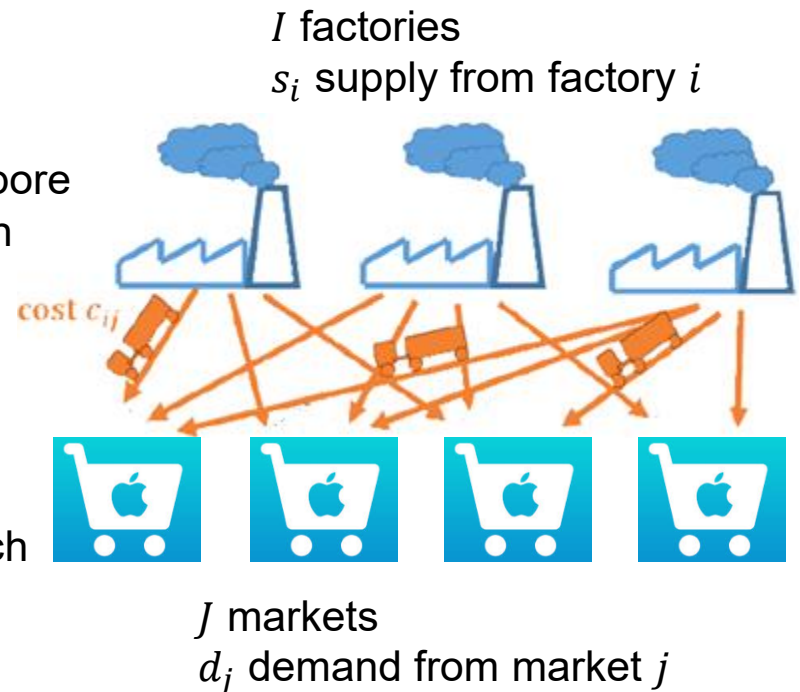
We want to decide how much to transport from each factory to each market in order to:

minimize the total transportation cost,

subject to: *Supply constraints:* The total amount shipped from each factory must not exceed its supply.

Demand constraints: The total amount received by each market must meet its demand.

Non-negativity: Quantities shipped must be greater than or equal to zero.



The maximum flow problem

A network of nodes and links, each with a capacity (e.g., bandwidth limit).

A source sends data, a sink receives it.

Intermediate nodes (e.g., routers) must forward data without storing it.

Goal:

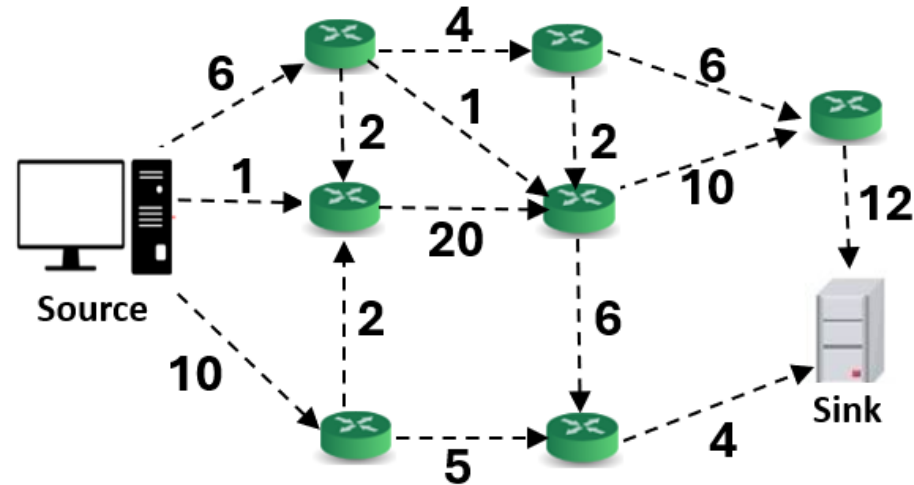
Maximize total data flow from source to sink

subject to:

Capacity limits on each link

Flow conservation at all intermediate nodes (flow in = flow out)

Non-negative flows



Resource allocation problem 1

Maximize academic benefit by selecting the best course combination within limits.

- Choose courses within a maximum credit limit (20 credits)
- Stay within a weekly workload limit (40 hours)
- Avoid time slot conflicts
- Select only whole courses (integer decision)

Decision variables:

$x_i = 1$ if course i is selected, 0 otherwise

Objective Function:

Maximize $Z = b_1x_1 + b_2x_2 + \dots + b_nx_n$ (b_i : benefit score of course i)

subject to:

- **Total credits:** $c_1x_1 + c_2x_2 + \dots + c_nx_n \leq 20$
- **Total workload:** $w_1x_1 + w_2x_2 + \dots + w_nx_n \leq 40$
- **No time conflicts:** $x_i + x_j \leq 1$ for overlapping courses
- $x_i \in \{0, 1\}$



Course	Credits	Workload per week	Time Slot	Benefit score
EE2213	4	10 hrs	Mon 10–12	9
MA2101	4	9 hrs	Mon 10–12	7
CDE2212	4	11 hrs	Tue 2–5	8
CE3201	2	5 hrs	Wed 3–5	5
EE3801	3	7 hrs	Thu 1–3	6

Problem formulation

x_1, x_2 : Decision variables

Maximize $150x_1 + 200x_2$ ← Objective function

subject to:

$$\begin{aligned}x_1 + x_2 &\leq 200 \\9x_1 + 3x_2 &\leq 1240 \\12x_1 + 16x_2 &\leq 2660 \\x_1, x_2 &\geq 0\end{aligned}$$

← Constraints

In linear program: **objective function** + **constraints** are **all linear**

Typically (not always): **variables are non-negative**

Linear programming uses only **non-strict inequalities** ($\leq, \geq$) and equalities ($=$), **not strict inequalities** ($<, >$)

If variables are integer: system called Integer Linear Programme (ILP)*.

* ILP involves more advanced mathematical complexity, so we won't cover it in lectures, but you'll gain hands-on experience with small-scale integer programs during tutorials.

Agenda

- Background of linear programs
- **Recap of linear algebra**
- Solving linear programs (LP)
 - The graphical method
 - Different cases of LP outcomes
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Some linear algebra review

Concept	Definition	Notation	Dimension	Key characteristics
Scalar	A single numerical value	A single number (e.g., 5, -2.3, π)	1×1	- A fundamental, indivisible quantity. - Has magnitude only.
Variable	A symbol representing a quantity that can change or take on different numerical values.	A letter (e.g., x, y, z, w_1)	1×1	- Represents a point in space, a direction, or a collection of features. - Has both magnitude and direction. - A 1D array of scalars.
Vector	An ordered list (array) of numbers (scalars).	Bold lowercase letters (e.g., $\mathbf{x}, \mathbf{y}$), or with an arrow ($\vec{\mathbf{x}}$); written as a row or column (e.g., $\mathbf{x} = [2, -1, 4]$)	$n \times 1$ or $1 \times n$	- Represents a point in space, a direction, or a collection of features. - Has both magnitude and direction. - A 1D array of scalars.
Matrix	A rectangular array of numbers (scalars) arranged in rows and columns.	Bold uppercase letters (e.g., $\mathbf{A}, \mathbf{M}$), e.g., $\mathbf{X} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	$m \times n$	- Organizes multi-dimensional data. - Can represent linear transformations or systems of equations. - A 2D array of scalars.

Capital Sigma: $\sum_{i=1}^n x_i = x_1 + x_2 + \cdots + x_{n-1} + x_n$

Capital Pi: $\prod_{i=1}^n x_i = x_1 \cdot x_2 \cdot \cdots \cdot x_{n-1} \cdot x_n$

Reference: <http://www.cs.cmu.edu/~zkolter/course/linalg>

Some linear algebra review

Vector or matrix **transpose**:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{x}^T = [x_1 \ x_2 \ \cdots \ x_n]$$

$$\mathbf{X} = \begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \cdots & x_{mn} \end{bmatrix}, \quad \mathbf{X}^T = \begin{bmatrix} x_{11} & \cdots & x_{m1} \\ \vdots & \ddots & \vdots \\ x_{1n} & \cdots & x_{mn} \end{bmatrix}$$

➤ Example:

$$\mathbf{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} \\ x_{2,1} & x_{2,2} & x_{2,3} \\ x_{3,1} & x_{3,2} & x_{3,3} \end{bmatrix}, \quad \mathbf{X}^T = \begin{bmatrix} x_{1,1} & x_{2,1} & x_{3,1} \\ x_{1,2} & x_{2,2} & x_{3,2} \\ x_{1,3} & x_{2,3} & x_{3,3} \end{bmatrix}$$

Some linear algebra review

Addition and subtraction of matrices and vectors is defined just as addition or subtraction of the elements, for $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$,

$$\mathbf{C} \in \mathbb{R}^{m \times n} = \mathbf{A} + \mathbf{B} \Leftrightarrow c_{ij} = a_{ij} + b_{ij}$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

➤ Example:

$$\begin{bmatrix} 2 & 8 \\ 3 & 7 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 2 + 1 & 8 + 2 \\ 3 + 0 & 7 + 5 \end{bmatrix} = \begin{bmatrix} 3 & 10 \\ 3 & 12 \end{bmatrix}$$

Some linear algebra review

Matrices multiplication: for two matrices $\mathbf{A} \in \mathbb{R}^{m \times d}$, $\mathbf{B} \in \mathbb{R}^{d \times h}$,

Dimension
must match

$$\mathbf{C} \in \mathbb{R}^{m \times h} = \mathbf{AB} \Leftrightarrow c_{ij} = \sum_{k=1}^d a_{ik} b_{kj}$$

➤ Example:

$$\mathbf{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}, \mathbf{B} = \begin{bmatrix} j & k & l \\ m & n & o \\ p & q & r \end{bmatrix}$$

$$\mathbf{A} \mathbf{B} = \begin{bmatrix} aj + bm + cp & ak + bn + cq & al + bo + cr \\ dj + em + fp & dk + en + fq & dl + eo + fr \\ gj + hm + ip & gk + hn + iq & gl + ho + ir \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}_{(2 \times 2)} \begin{bmatrix} 3 & 1 & 5 \\ -2 & 1 & 3 \end{bmatrix}_{(2 \times 3)} = \begin{bmatrix} & & \\ & & \end{bmatrix}_{(2 \times 3)}$$

Source:

<https://notesbylex.com/matrix-multiplication>

Some linear algebra review

Special case of matrices multiplication: **inner product** of two vectors,
for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i$$

➤ Example: $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$
 $\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + x_2 y_2$

Special case: product of matrix and vector, for $\mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{x} \in \mathbb{R}^n$,

$$\mathbf{Ax} \in \mathbb{R}^m = \begin{bmatrix} a_1^T \mathbf{x} \\ \vdots \\ a_m^T \mathbf{x} \end{bmatrix} = \sum_{i=1}^n a_i x_i$$

➤ Example:
$$\begin{pmatrix} a & b & c \\ l & m & n \\ r & s & t \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} \longrightarrow \begin{pmatrix} \\ \\ \end{pmatrix}$$

Source:

https://phys.libretexts.org/Courses/University_of_California_Davis/UCD%3A_Physics_9HB__Special_Relativity_and_Thermal_Statistical_Physics/3%3A_Spacetime/3.1%3A_Vector_Rotations

Some linear algebra review

The **identity matrix** $\mathbf{I} \in \mathbb{R}^{n \times n}$ is a **square matrix** with:

- 1s on the diagonal
- 0s elsewhere

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

It acts like “1” in matrix multiplication, *i.e.*, for any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$:

$$\mathbf{I}\mathbf{A} = \mathbf{A} \quad (\text{with } \mathbf{I} \in \mathbb{R}^{m \times m})$$

$$\mathbf{A}\mathbf{I} = \mathbf{A} \quad (\text{with } \mathbf{I} \in \mathbb{R}^{n \times n})$$

For a **square matrix** $\mathbf{A} \in \mathbb{R}^{n \times n}$, its **inverse matrix** $\mathbf{A}^{-1}$ is the matrix such that,

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{I} = \mathbf{A}\mathbf{A}^{-1}$$

Not all matrices have an inverse (they must be **square** and **non-singular**).

Linear function

A function $f: \mathcal{R}^n \rightarrow \mathcal{R}$ is **linear** if it can be written as a weighted sum of variables:

$$f(\mathbf{x}) = a_1x_1 + a_2x_2 + \cdots + a_nx_n = \mathbf{a}^T \mathbf{x}$$

➤ **Example:** $f(x_1, x_2) = 2x_1 - 7x_2 = \begin{bmatrix} 2 \\ -7 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

This means it contains no products, no powers, and no nonlinear functions of variables.

Mathematically, this form arises because linear functions satisfy:

- **Homogeneity (Scaling):** $f(\lambda \mathbf{x}) = \lambda f(\mathbf{x})$
- **Additivity (Adding):** $f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$

Linear vs affine function

A function $f: \mathcal{R}^n \rightarrow \mathcal{R}$ is **affine** if and only if it can be written as:

$$f(\mathbf{x}) = \mathbf{a}^T \mathbf{x} + b$$

where $\mathbf{a}, \mathbf{x} \in \mathcal{R}^n, b \in \mathcal{R}$

Here, b is a scalar, often referred to as the **offset** or **bias**.

➤ **Example:** $f(x_1, x_2) = 2x_1 - 7x_2 + 6 = \begin{bmatrix} 2 \\ -7 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + 6$

An affine function is a linear function with an added constant term.

Linear vs affine function

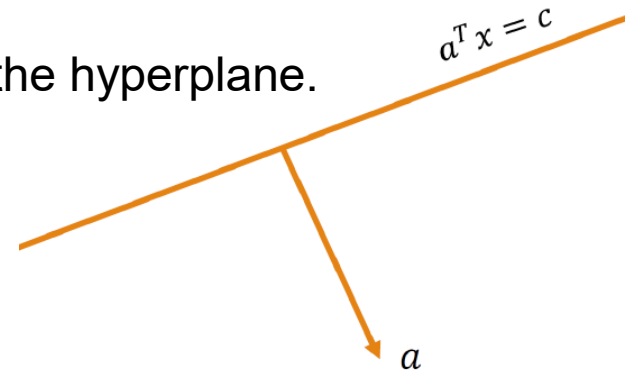
An **affine function** is **not necessarily a linear function**, but every linear function is a special case of an affine function.

An **affine function** $f(x) = \mathbf{a}^T \mathbf{x} + b$ is a **linear function** if and only if the constant term $b = 0$.

Linear function: Geometric interpretation

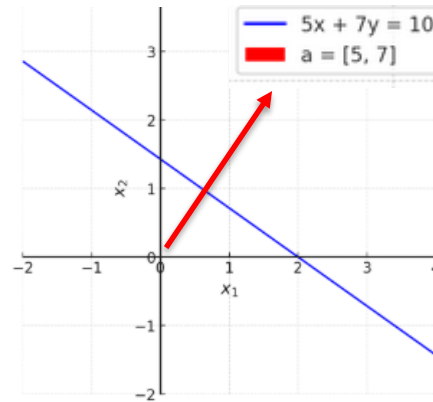
Geometric interpretation of $\mathbf{a}^T \mathbf{x} = c$

- The equation $\mathbf{a}^T \mathbf{x} = c$ represents a **hyperplane** in $\mathcal{R}^n$ (a line in $\mathcal{R}^2$, a plane in $\mathcal{R}^3$, etc.)
- The vector $\mathbf{a}$ is a normal (perpendicular) to the hyperplane.



- In 2D: $\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ is a perpendicular to the line $a_1 x_1 + a_2 x_2 = c$

➤ **Example:** $5x_1 + 7x_2 = 10$

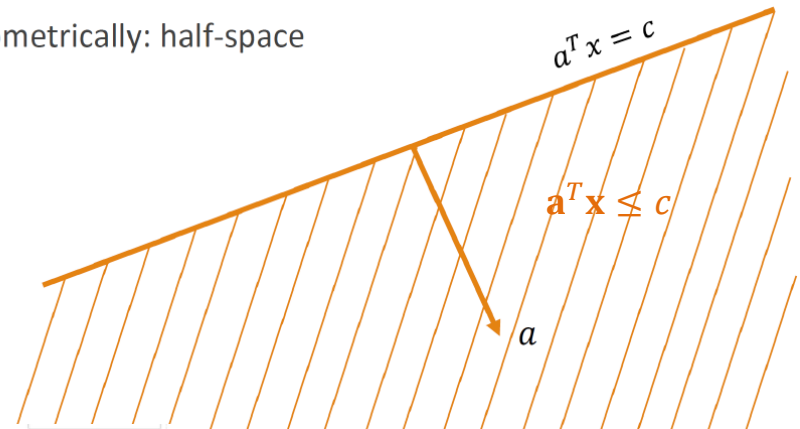


Linear inequality

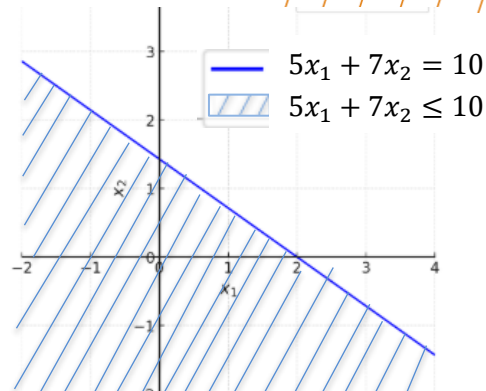
The inequality $\mathbf{a}^T \mathbf{x} \leq c$ defines a **half-space** in geometry.

It includes all points x that lie on or below the hyperplane $\mathbf{a}^T \mathbf{x} = c$

Geometrically: half-space



➤ Example: $5x_1 + 7x_2 \leq 10$

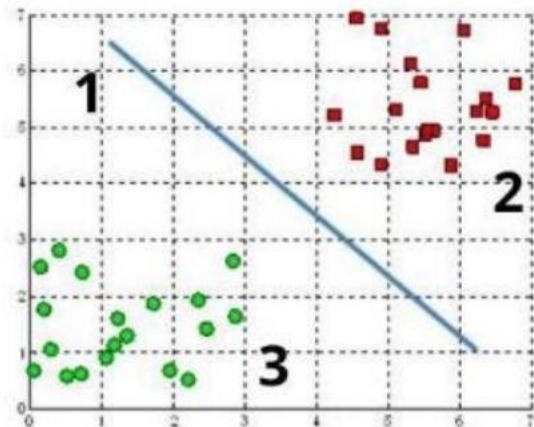


Visualize in higher-dimensional space

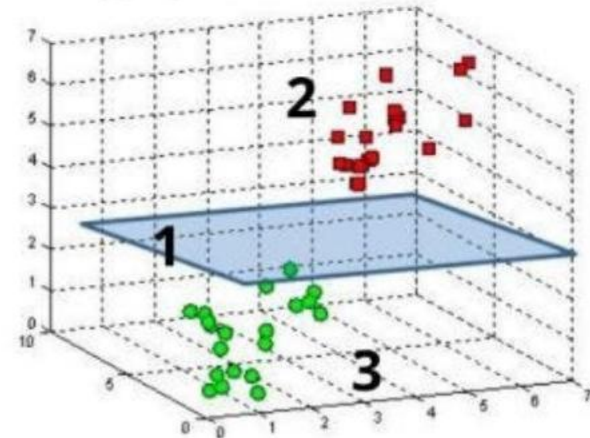
Hyperplane: The region defined by a linear equality $a_i x = b_i$

Half-space: The region defined by a linear inequality $a_i x \leq b_i$

1	$\beta_o + \beta_1 X_1 + \beta_2 X_2 = 0$	$\rightarrow$	hyperplane	$\leftarrow$	1	$\beta_o + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 = 0$
2	$\beta_o + \beta_1 X_1 + \beta_2 X_2 > 0$	$\searrow$	half-space	$\swarrow$	2	$\beta_o + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 > 0$
3	$\beta_o + \beta_1 X_1 + \beta_2 X_2 < 0$	$\nearrow$		$\nwarrow$	3	$\beta_o + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 < 0$



R2 Space, Hyperplane is a line



R3 Space, Hyperplane is a plane

Credit: <https://www.enjoyalgorithms.com/blog/support-vector-machine-in-ml>

Linear Program (LP) Terminology

Linear programs can be written under the standard form:

Maximize

$$c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

Subject to

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \leq b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \leq b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \leq b_m$$

$$x_1, x_2, \dots, x_n \geq 0$$

Concisely written as:

Maximize

$$\mathbf{c}^T \mathbf{x}$$

Subject to

$$\mathbf{Ax} \leq \mathbf{b}$$

$$\mathbf{x} \geq 0$$

$$\mathbf{x} = [x_1, x_2 \cdots x_n]^T$$

$$\mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{b} \in \mathbb{R}^{m \times 1}, \mathbf{c} \in \mathbb{R}^{n \times 1}$$

Linear Program (LP) Terminology

In matrix form, a linear program in standard form can be written as:

$$\text{Maximize} \quad \mathbf{c}^T \mathbf{x}$$

$$\begin{aligned} \text{Subject to} \quad & \mathbf{Ax} \leq \mathbf{b} \\ & \mathbf{x} \geq 0 \end{aligned}$$

where $\mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$, and $\mathbf{A} = [a_{ij}]$ is the $m \times n$ matrix
whose i, j element is a_{ij}

Quick question 1

Select all options that are **linear programs**.

(A) Minimize $3x + 4y$

$$\begin{aligned} \text{s.t.} \quad & x + 2y \geq 8 \\ & 5x + y = 10 \\ & x, y \geq 0 \end{aligned}$$

(B) Maximize $7u + 9v$

$$\begin{aligned} \text{s.t.} \quad & 2u - v \leq 6 \\ & uv \leq 10 \\ & u, v \geq 0 \end{aligned}$$

(C) Maximize $5p + q$

$$\begin{aligned} \text{s.t.} \quad & 3p + 2q < 12 \\ & p, q \geq 0 \end{aligned}$$

(D) Minimize $4a - b$

$$\begin{aligned} \text{s.t.} \quad & -a + 3b \geq 9 \\ & 2a + b \leq 7 \\ & a, b \geq 0 \end{aligned}$$

Agenda

- Background of linear programs
- Recap of linear algebra
- **Solving linear programs (LP)**
 - **The graphical method**
 - Different cases of LP outcomes
 - The Simplex algorithm

Graphical method

For LPs with only two decision variables, we often solve them using the **graphical method**.

Consider the bakery production example:

$$\begin{array}{ll}\text{Maximize} & x_1 - 2x_2 \\ \text{Subject to} & x_1 \geq 0 \\ & x_2 \geq 0 \\ & 3x_1 + 2x_2 \leq 6 \\ & 2x_1 + 3x_2 \leq 6\end{array}$$

Graphical method

Maximize $x_1 - 2x_2$

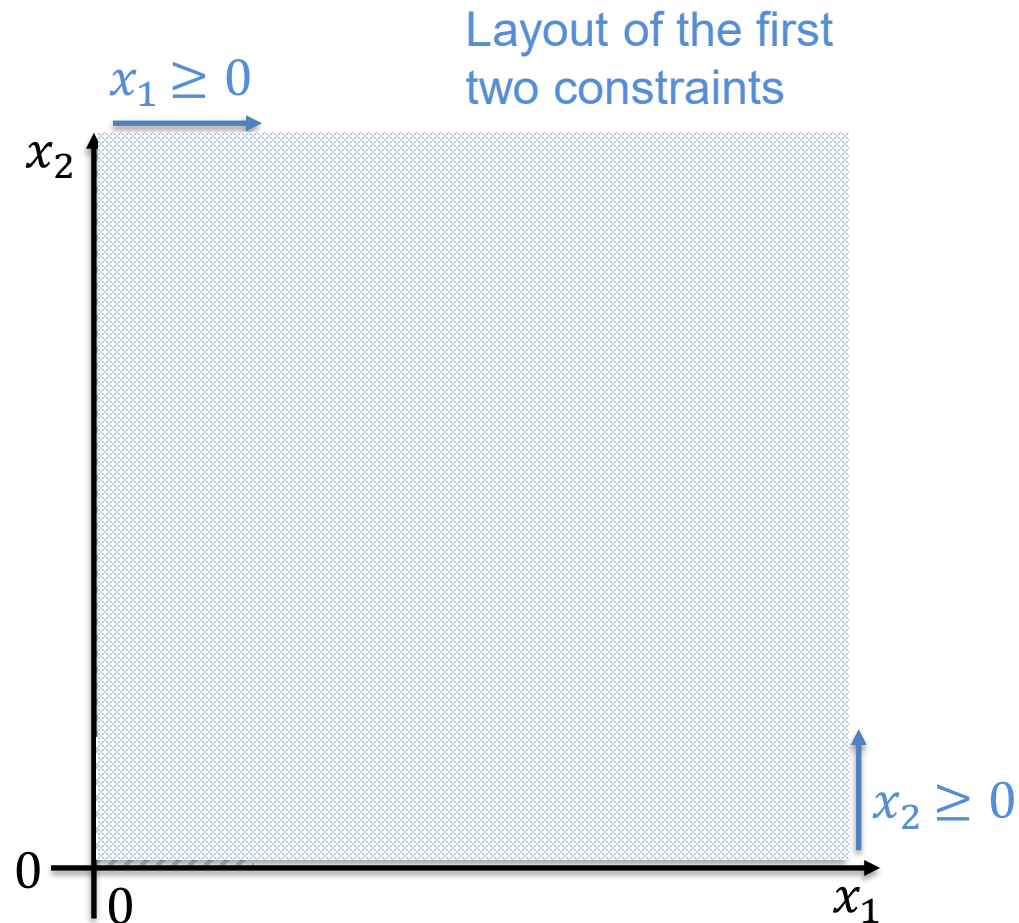
Subject to

$$x_1 \geq 0$$

$$x_2 \geq 0$$

$$3x_1 + 2x_2 \leq 6$$

$$2x_1 + 3x_2 \leq 6$$



Graphical method

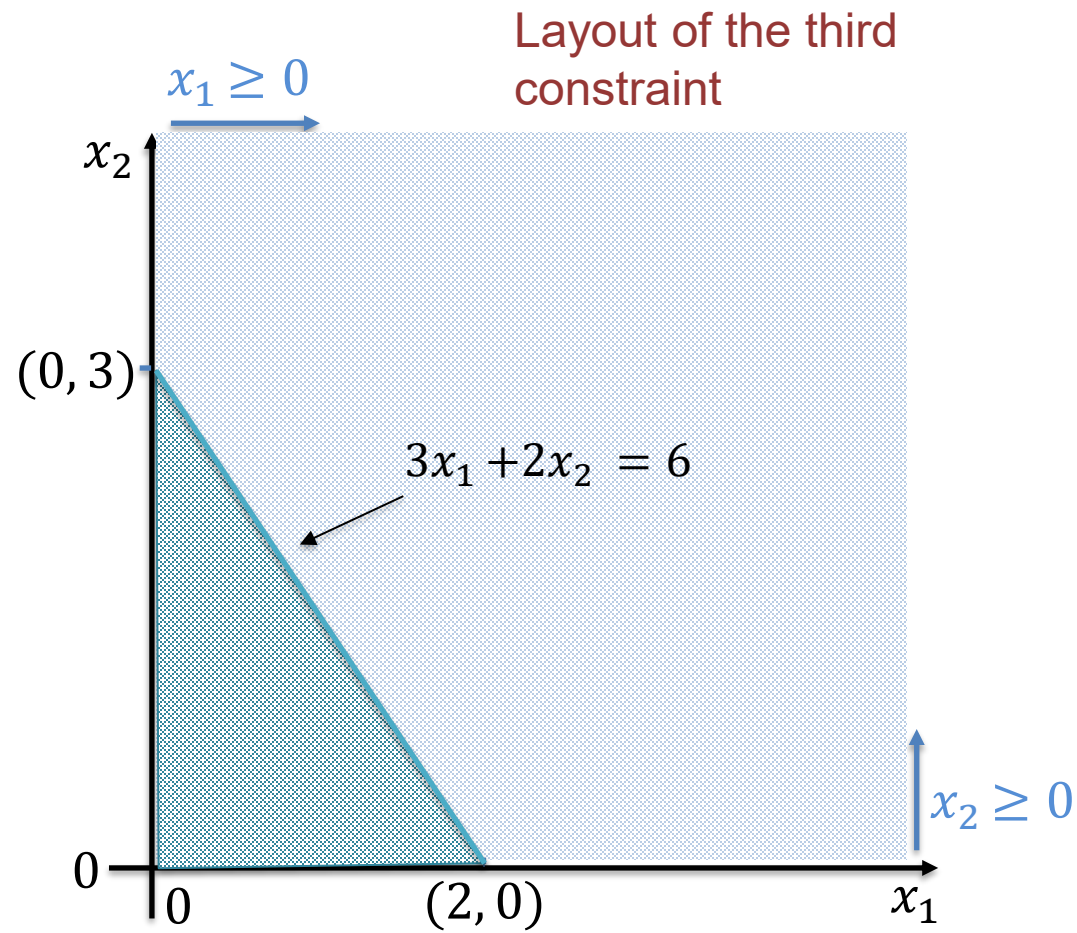
Maximize $x_1 - 2x_2$

Subject to $x_1 \geq 0$

$x_2 \geq 0$

→ $3x_1 + 2x_2 \leq 6$

$2x_1 + 3x_2 \leq 6$



Graphical method

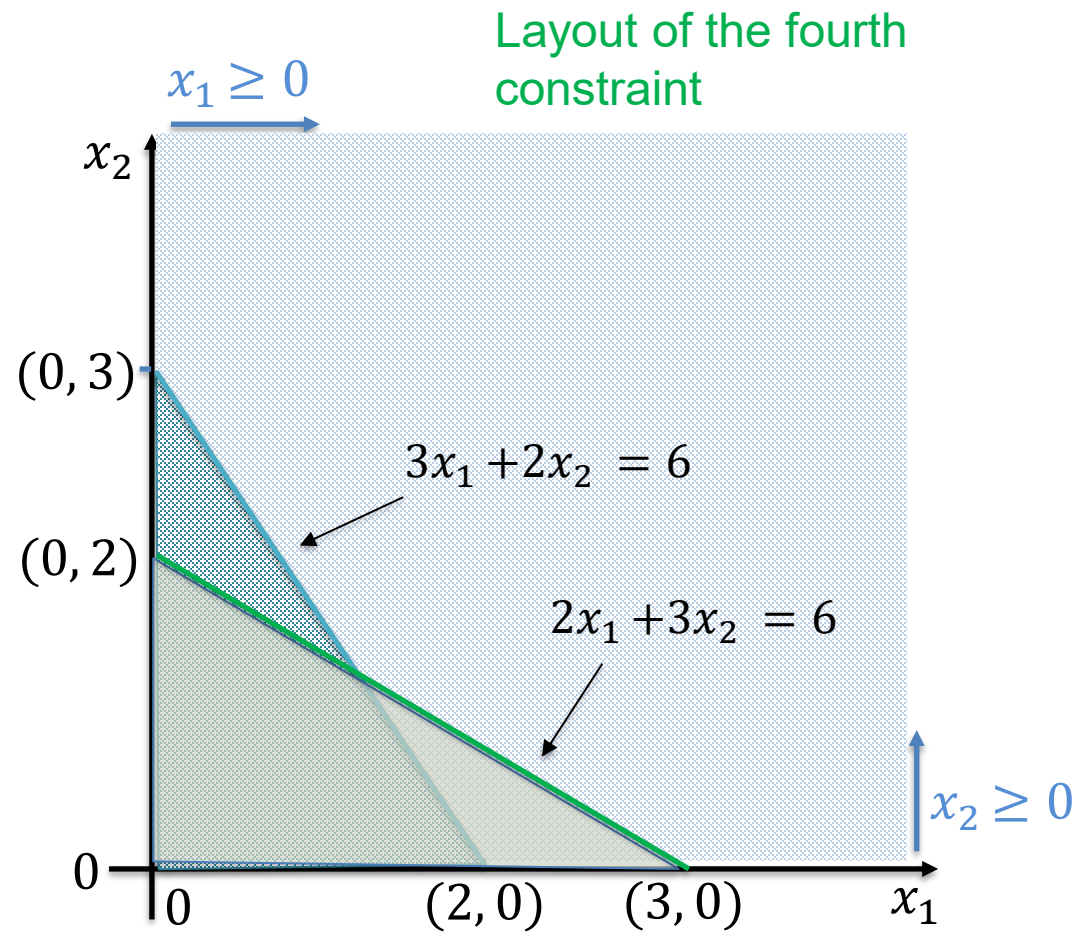
Maximize $x_1 - 2x_2$

Subject to $x_1 \geq 0$

$x_2 \geq 0$

$3x_1 + 2x_2 \leq 6$

→ $2x_1 + 3x_2 \leq 6$

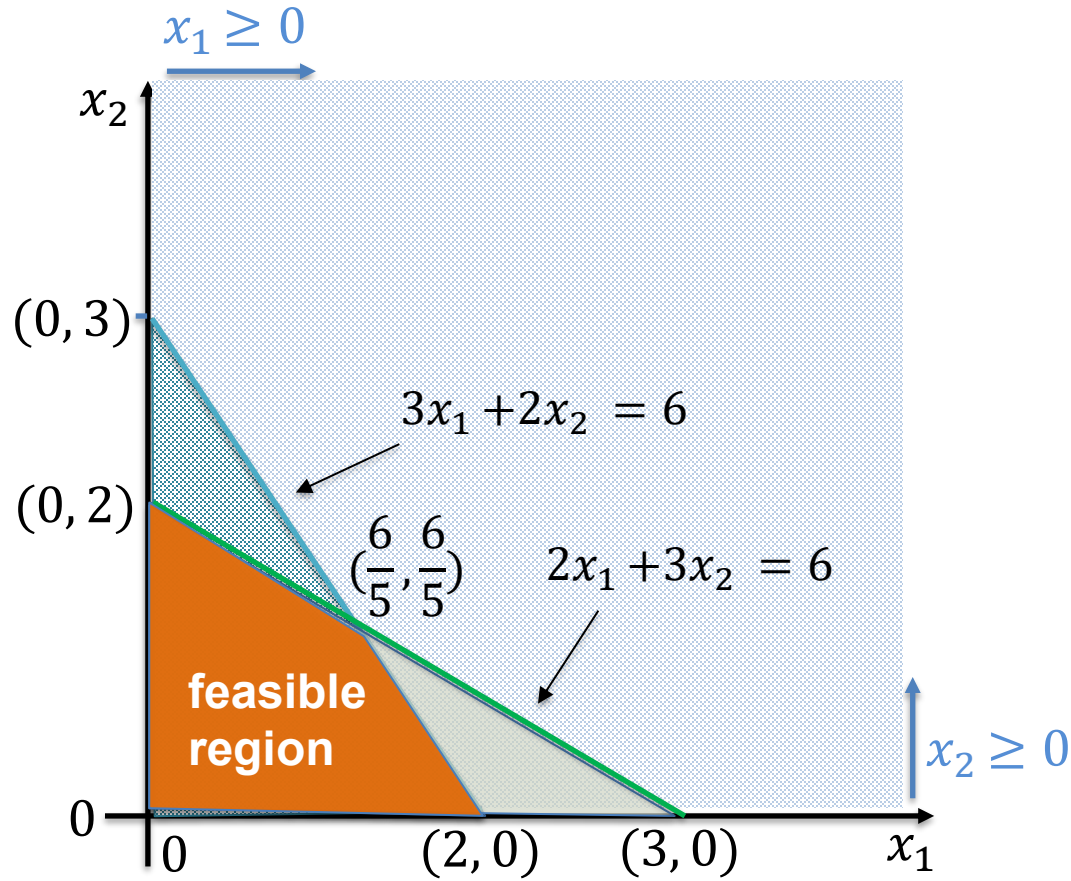


Graphical method

Maximize $x_1 - 2x_2$

Subject to $x_1 \geq 0$
 $x_2 \geq 0$
 $3x_1 + 2x_2 \leq 6$
 $2x_1 + 3x_2 \leq 6$

Feasible region: the set of all points that satisfy all the constraints of a LP problem.

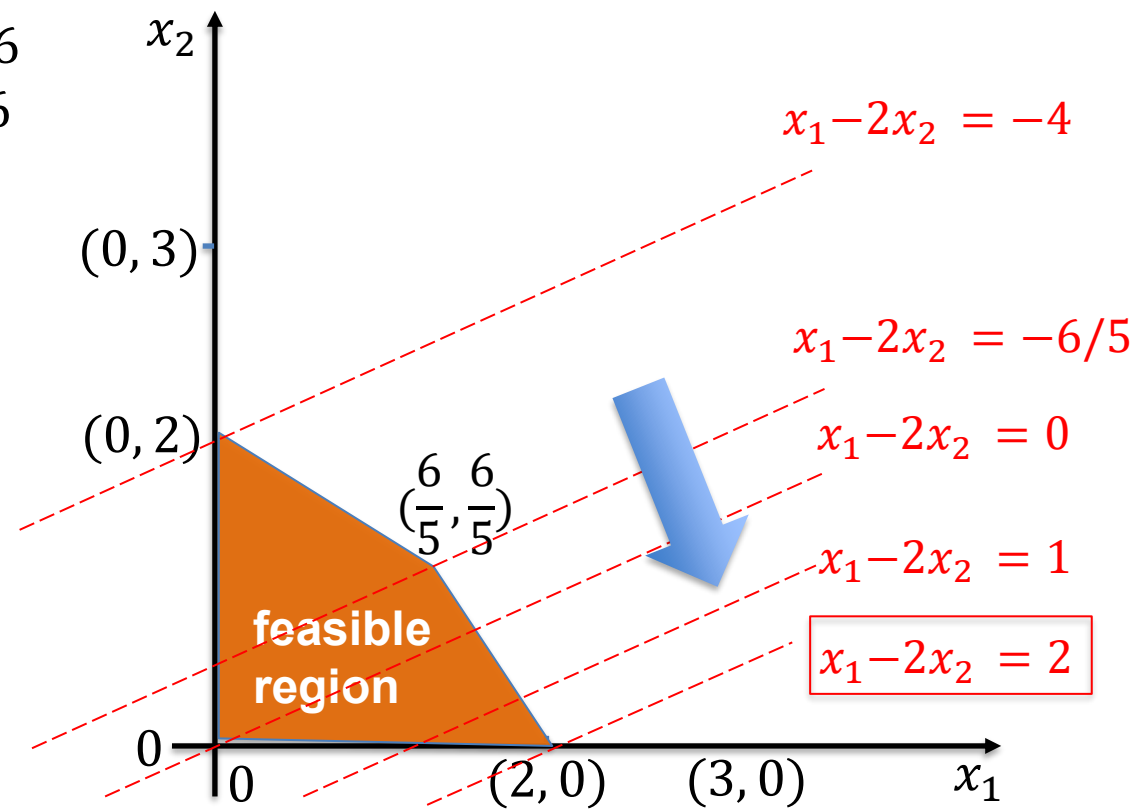


Graphical method

Maximize $x_1 - 2x_2$ 

Subject to
 $x_1 \geq 0$
 $x_2 \geq 0$
 $3x_1 + 2x_2 \leq 6$
 $2x_1 + 3x_2 \leq 6$

Objective = 2

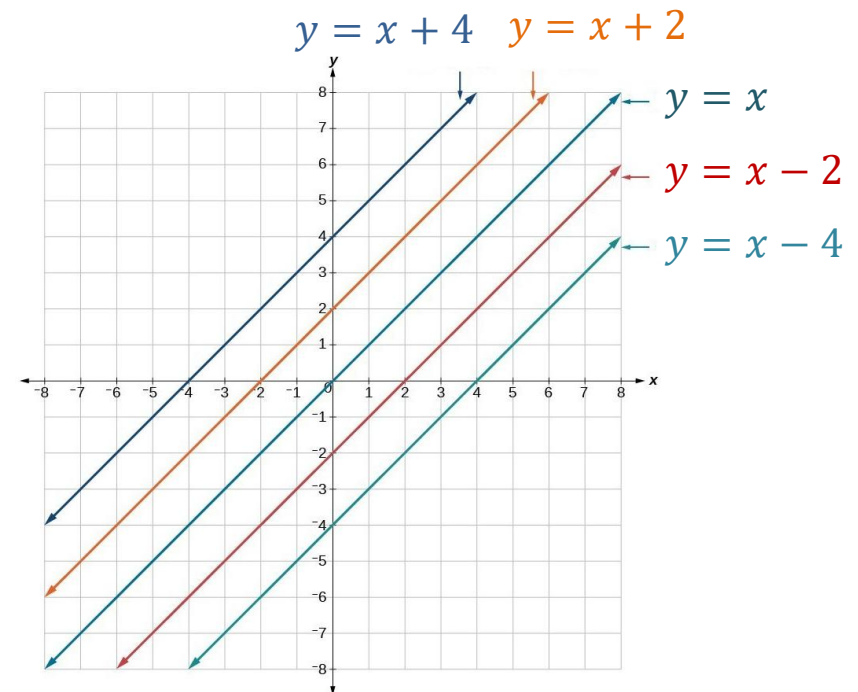
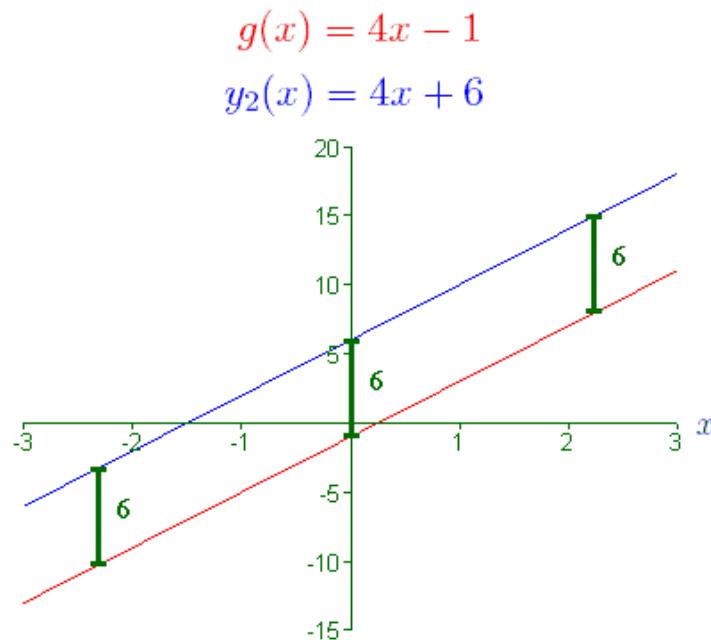


Recap: vertical and horizontal shift

graph: $y = f(x)$

$y = f(x) + k$ vertical shift by k units

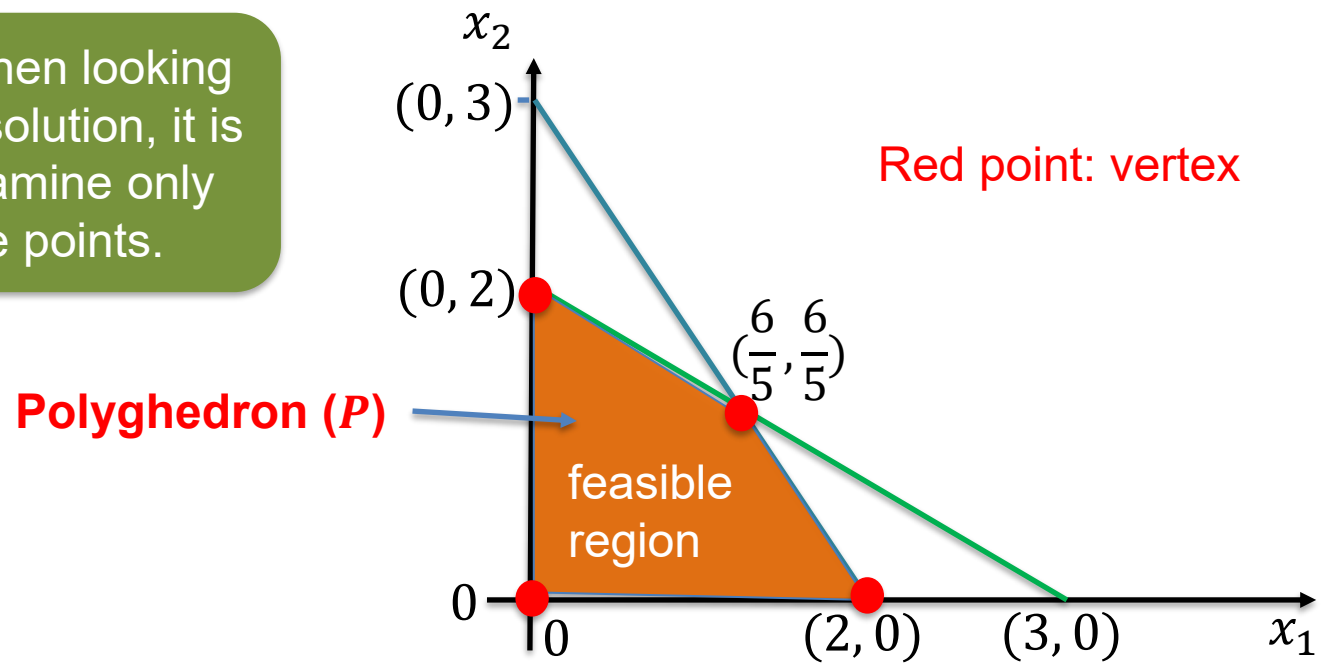
$y = f(x + h)$ horizontal shift by h units



Fundamental theorem of LP

If a linear program has an optimal solution, then there exists at least one optimal solution at a **vertex (corner point)** of the feasible region (or along an edge between vertices).

Implication: When looking for an optimal solution, it is enough to examine only the extreme points.

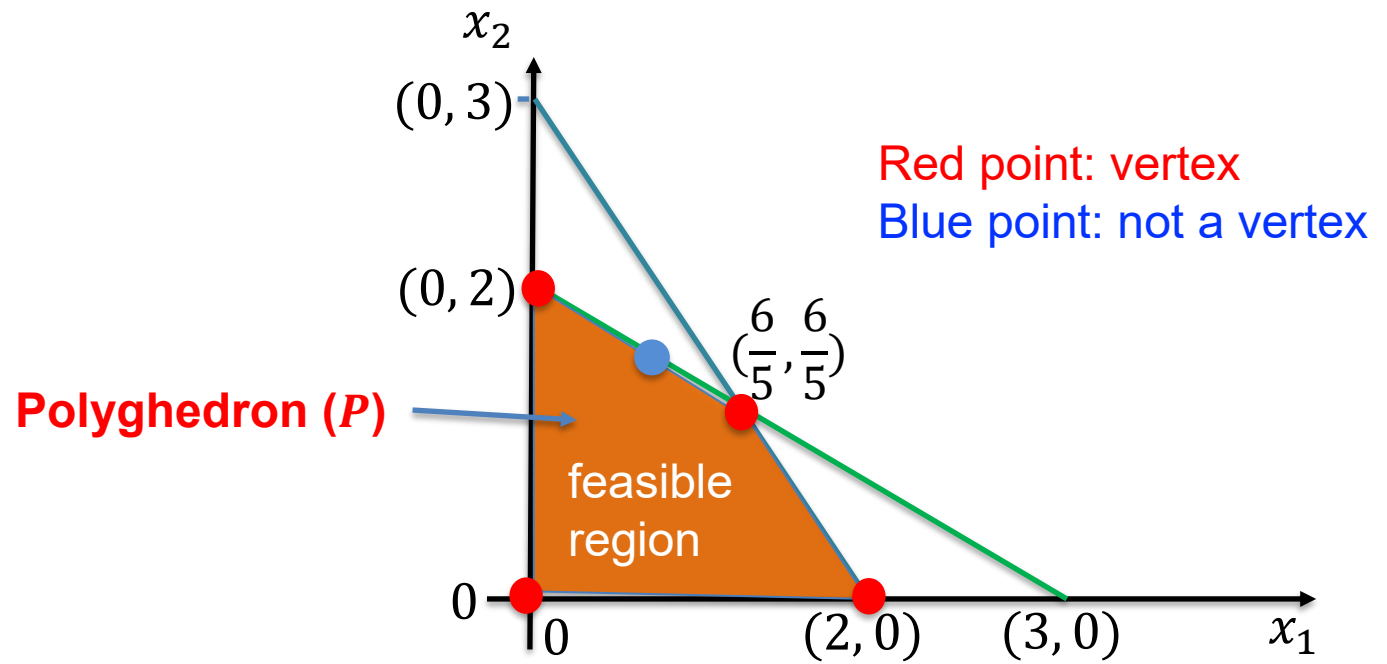


For proof, see
https://www.princeton.edu/~aaa/Public/Teaching/ORF363_COS323/F14/ORF363_COS323_F14_Lec11.pdf P13-P14

Terminology

A **vertex** (or **corner point**) is a point on the boundary of the feasible region where two or more constraints *intersect*; **the optimal solution of an LP (if it exists) occurs at one of these vertices.**

Polyhedron (P): the geometric shape formed by the intersection of all linear constraints; it represents the **feasible region** in LP



Summary of graphical method

- Works best for LPs with **two variables**.

Steps:

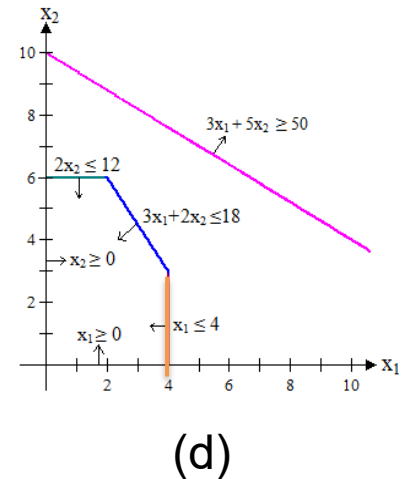
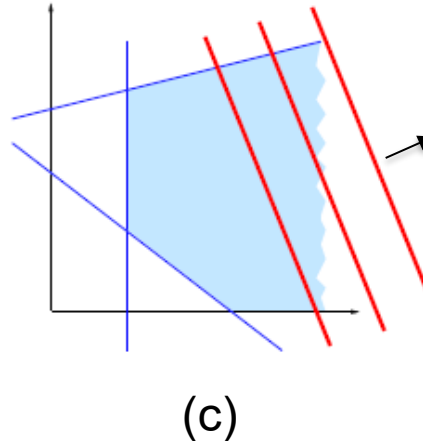
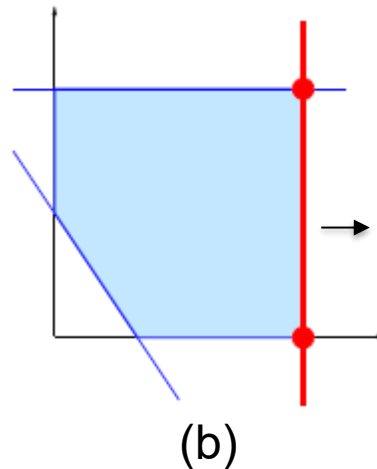
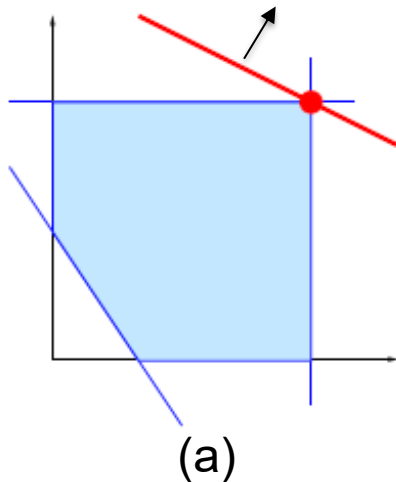
- Plot all constraints to identify the **feasible region**.
- Draw **objective function level lines** (parallel lines for different values).
- Move the level lines in the direction of optimization (maximize/minimize).
- **Optimal solution** occurs at a **vertex** (corner point) of the feasible region.

Limitations: Works best for 2 variables (easy to draw). For higher dimensions, use **Simplex or other algorithms**.

Agenda

- Background of linear programs
- Recap of linear algebra
- **Solving linear programs (LP)**
 - The graphical method
 - **Different cases of LP outcomes**
 - The Simplex algorithm

Quick question 2: Different cases of LP



For each case, consider:

- Does an optimal solution exist?
- If yes, is it unique?
- If not, why not?

Blue shaded area: feasible region
Red lines: objective function levels

Three types of LPs

Every LP falls into one of these cases:

Infeasible — No solution satisfies all constraints.

Unbounded — Objective can improve indefinitely within the feasible region.

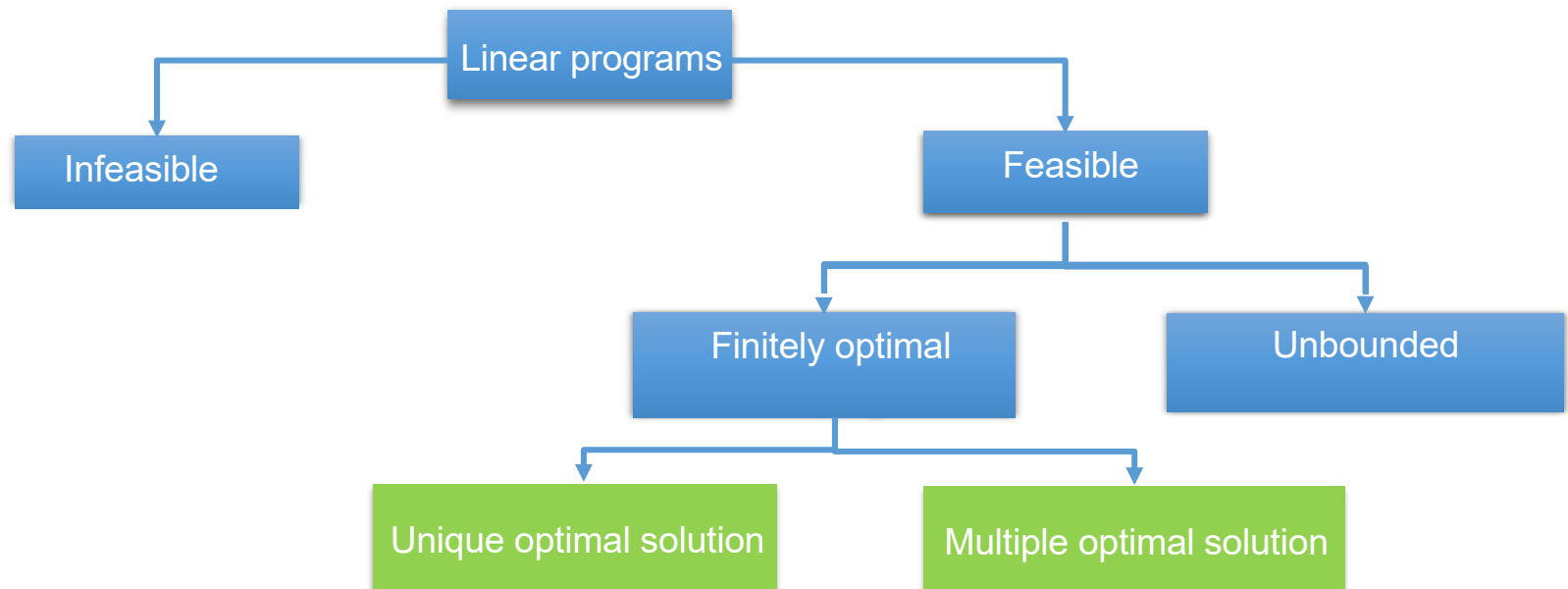
Finitely optimal — An optimal solution exists and is finite.

If finitely optimal, the solution can be:

Unique, or

Multiple optimal solutions along an edge or face of the feasible region.

Summary: Different cases of LP outcomes



Example: Infeasible problem

An LP is **infeasible** if its feasible region is empty.

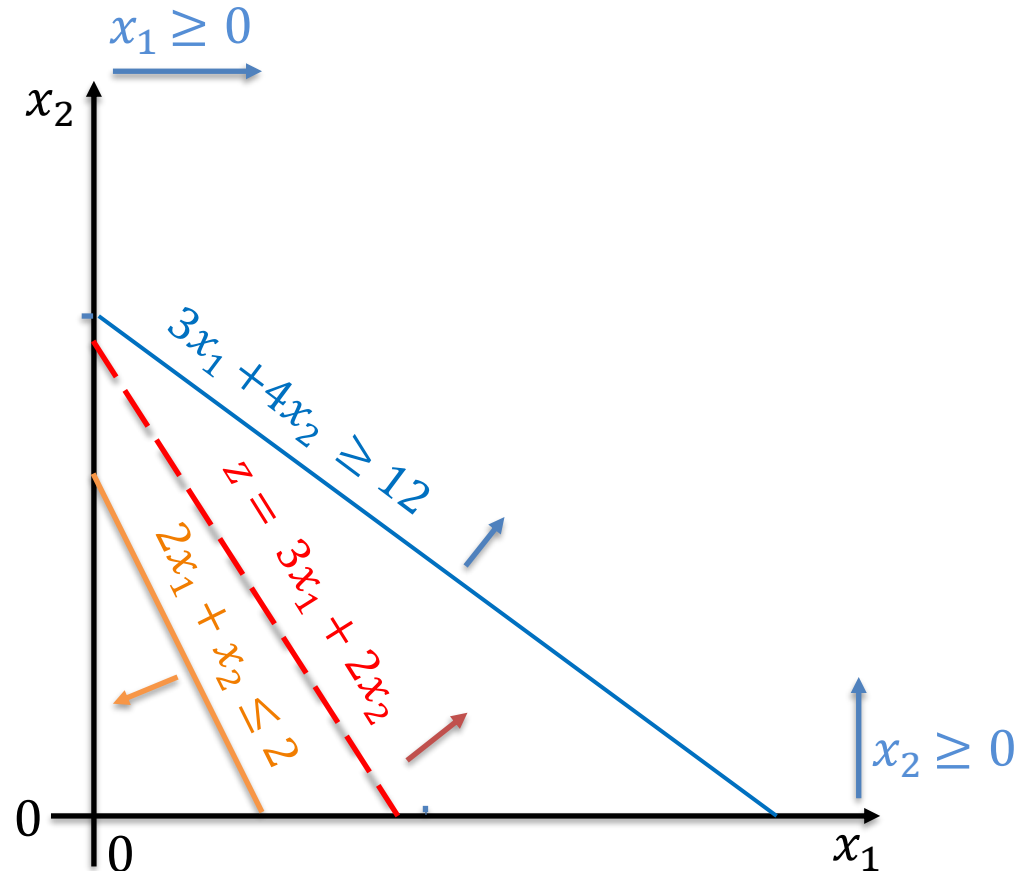
Maximize $z = 3x_1 + 2x_2$

subject to:

$$2x_1 + x_2 \leq 2$$

$$3x_1 + 4x_2 \geq 12$$

$$x_1, x_2 \geq 0$$



Try out yourself online: <https://www.desmos.com/calculator>

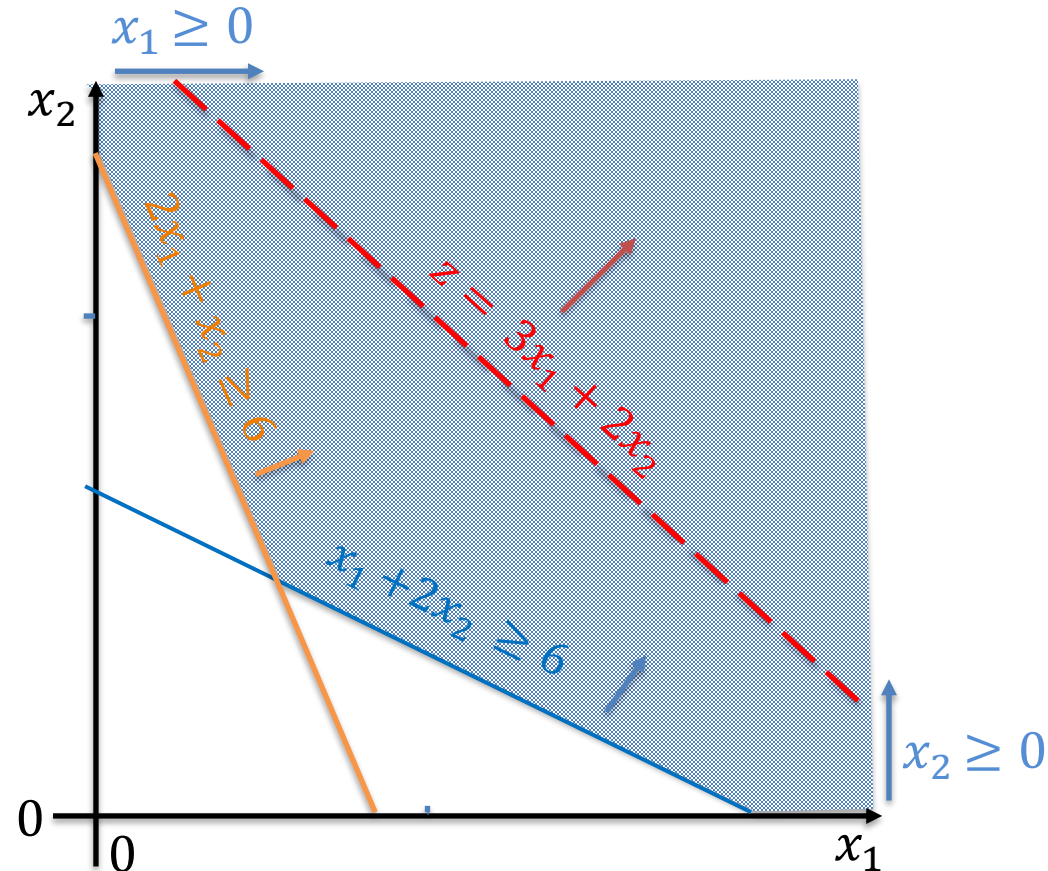
Example: Unbounded problem

An LP is **unbounded** if for any feasible solution, there is another feasible solution that is better.

Maximize $z = x_1 + x_2$

subject to:

$$\begin{aligned} x_1 + 2x_2 &\geq 6 \\ 2x_1 + x_2 &\geq 6 \\ x_1, x_2 &\geq 0 \end{aligned}$$



Example: Unbounded problem (cont.)

Clarify a common misconception:

An unbounded feasible region does *not* mean the LP is unbounded.

Minimize $z = x_1 + x_2$

subject to:

$$x_1 + 2x_2 \geq 6$$

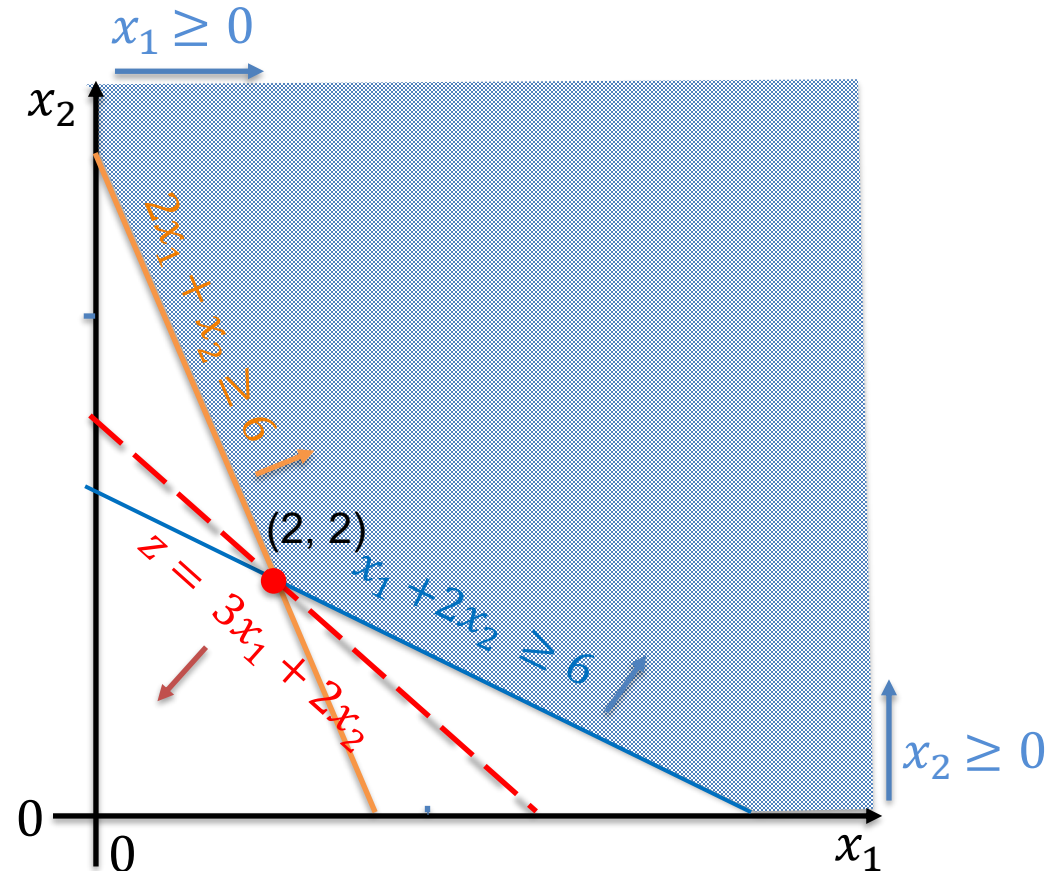
$$2x_1 + x_2 \geq 6$$

$$x_1, x_2 \geq 0$$

The optimal solution is:

$$x_1 = 2, \quad x_2 = 2$$

$$z_{\min} = 4$$



Example: Multiple optimal solution

An LP may have **multiple** optimal solutions.

Minimize $z = x_1 + 2x_2$

subject to:

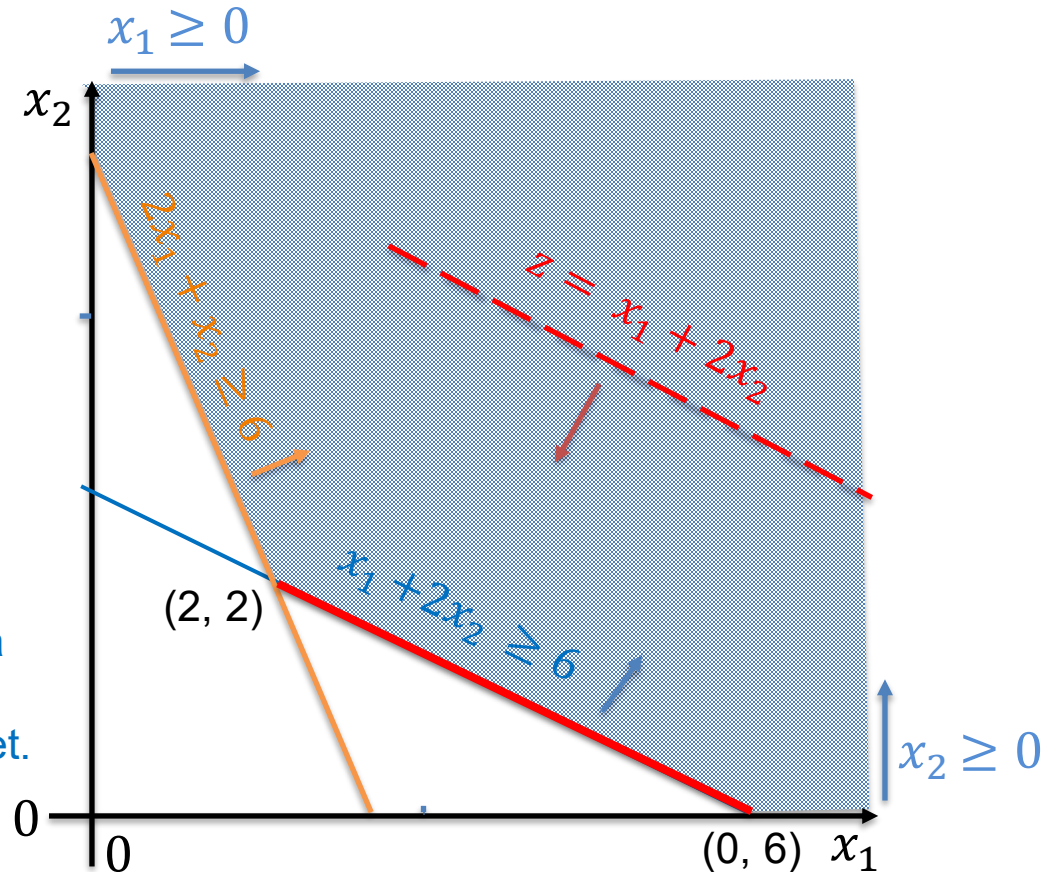
$$x_1 + 2x_2 \geq 6$$

$$2x_1 + x_2 \geq 6$$

$$x_1, x_2 \geq 0$$

$$z_{\min} = 6$$

Finitely Optimal: **Objective** reaches a finite maximum/minimum, even if optimal solutions form a continuous set.



Optimal solution vs. Optimal objective value

- **Optimal Solution:**

The value of the **decision variable(s)** that gives the best result for the objective function.

Tells us *where* the minimum or maximum occurs.

- **Optimal Objective Value:**

The *best value* achieved by the **objective function**.

Tells us *what* the minimum or maximum value is.

Optimal solution: $x^* = \operatorname{argmin}_{x \in \mathcal{F}} f(x)$

Optimal objective value $f^* = \min_{x \in \mathcal{F}} f(x) = f(x^*)$

➤ **Example:**

maximize x^2 , s.t. $x \in [-2, 2]$

Optimal solution: $-2, 2$

Optimal objective value 4

Quick question 3

Maximize $z = 2x_1 + 2x_2$

subject to:

$$x_1 - x_2 \geq 1$$

$$x_1 + x_2 \geq 3$$

$$x_1, x_2 \geq 0$$

Which of the following describes the feasible region and the LP?

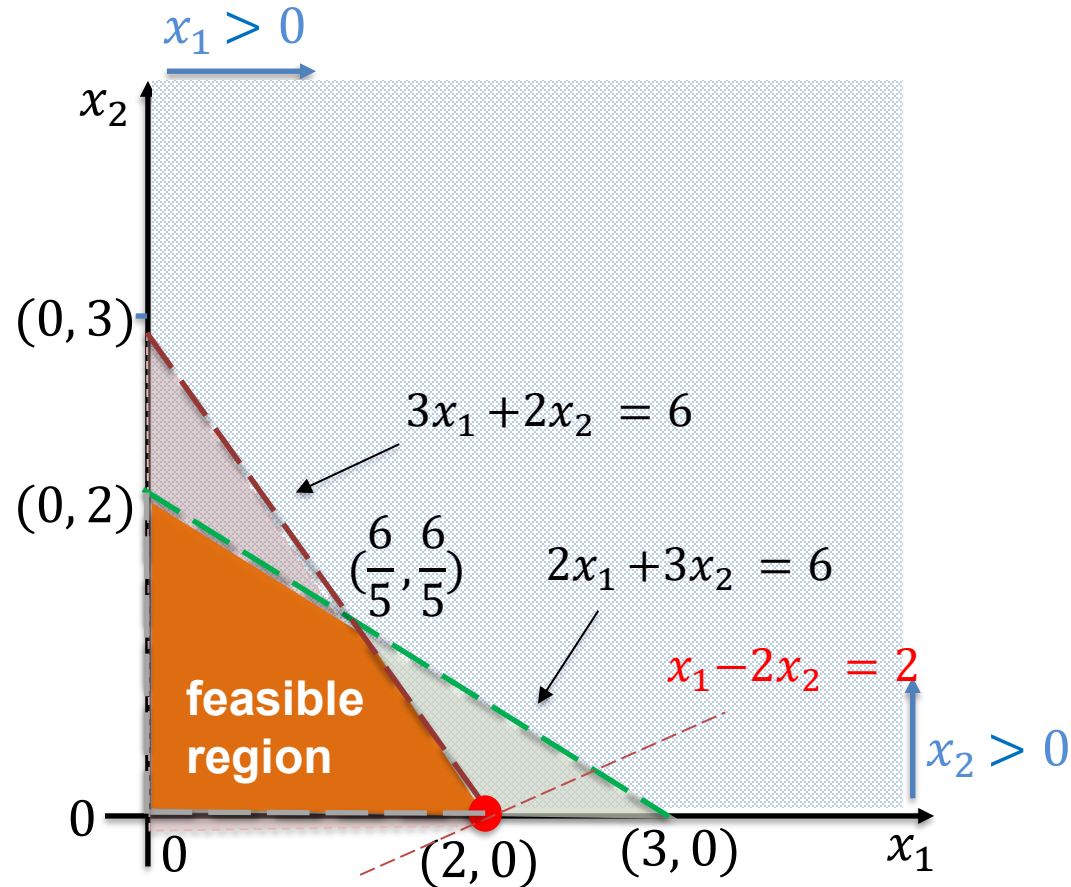
- (A) Bounded and feasible
- (B) Feasible but not bounded
- (C) Bounded but not feasible
- (D) Neither bounded nor feasible

Why not strict inequality?

Maximize $x_1 - 2x_2$

Subject to:
 $x_1 > 0$
 $x_2 > 0$
 $3x_1 + 2x_2 < 6$
 $2x_1 + 3x_2 < 6$

- The feasible set becomes open if strict inequalities are used.
- This means the boundary line is excluded from the feasible region.
- But in LP, the optimal solution lies on the boundary.



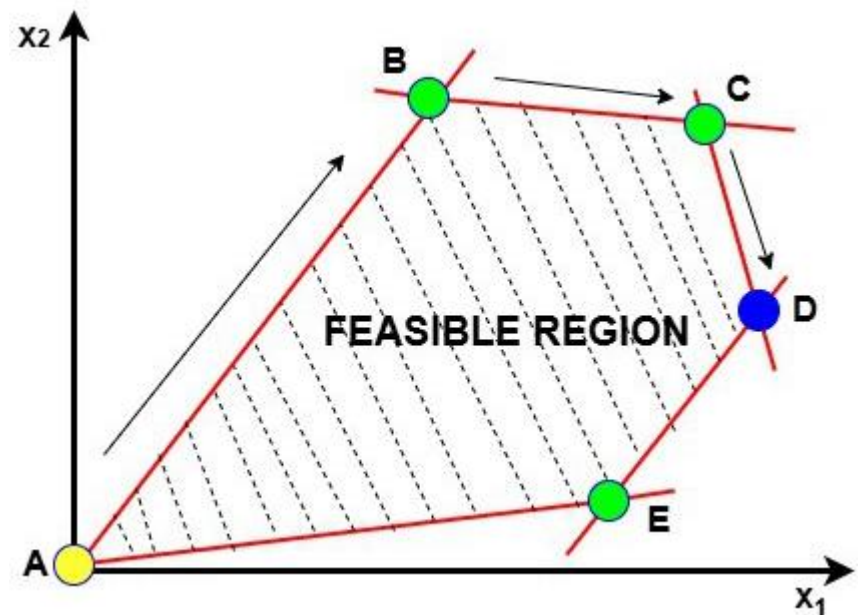
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Simplex algorithm

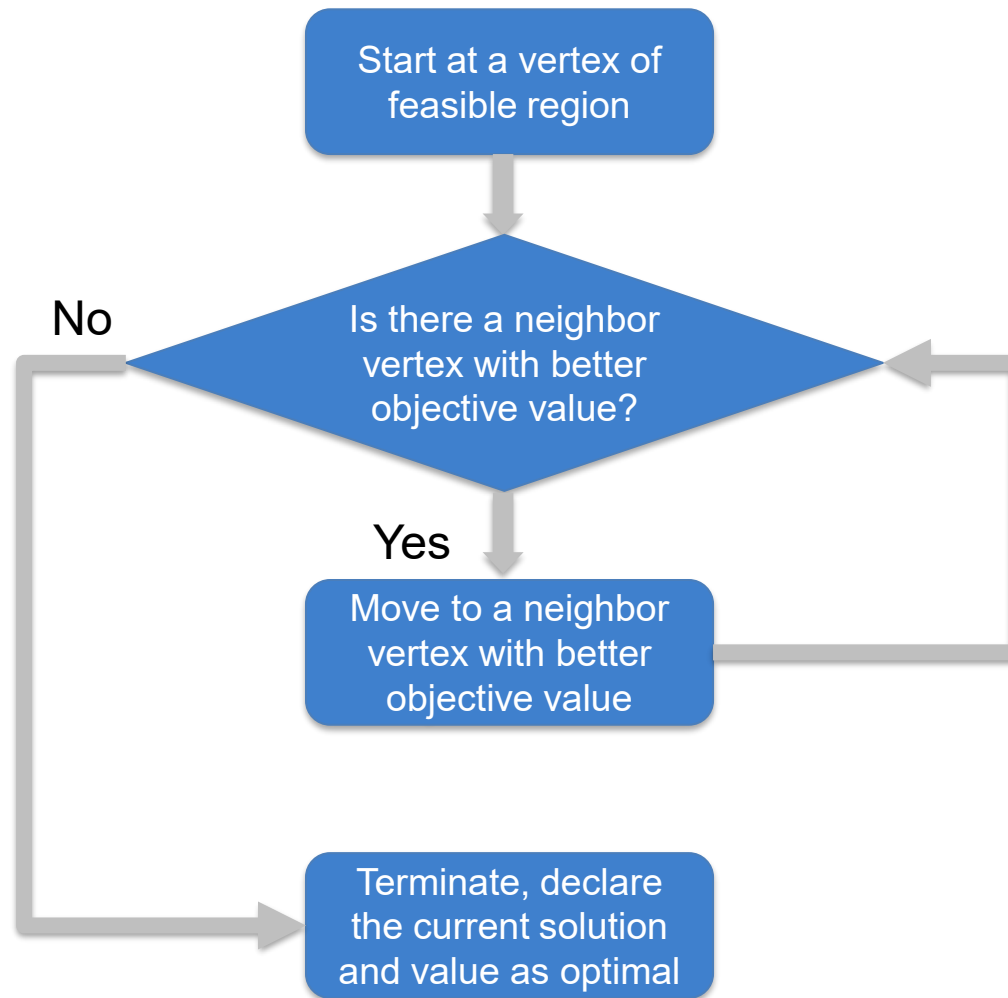
Basic idea:

Move from vertex to vertex along the edges of the feasible polytope, always in a direction that improves the objective function, until no further improvement is possible.



Simplex = *systematic exploration* of corners of the feasible region.
Relies on the same idea: **optimal solutions lie on vertices.**

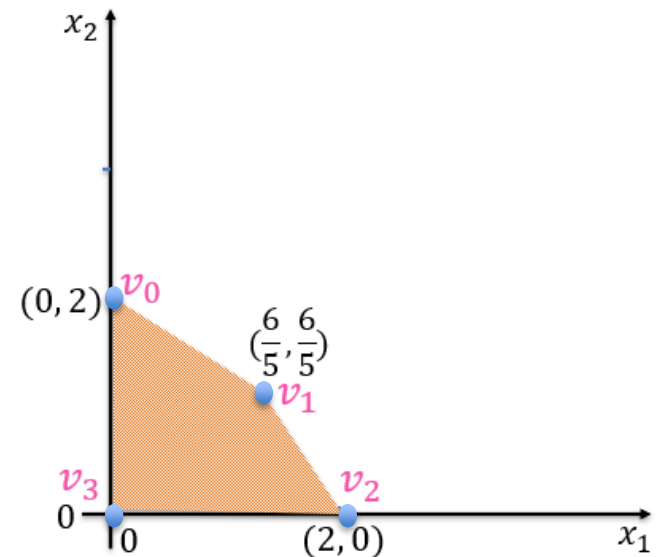
Simplex algorithm implementation



Simplex algorithm: Example

Maximize $x_1 - 2x_2$

Subject to $x_1 \geq 0$
 $x_2 \geq 0$
 $3x_1 + 2x_2 \leq 6$
 $2x_1 + 3x_2 \leq 6$



Step1: Evaluate objective

at v_0 : $z(v_0) = 0 - 4 = -4$

Step2: Identify adjacent vertices: v_1, v_3

Step 3: Evaluate objective at v_1 and v_3 :

$$z(v_1) = \frac{6}{5} - 2 * \frac{6}{5} = -1.2; \quad z(v_3) = 0$$

So move to the adjacent vertex with largest improvement. $v_0 \rightarrow v_3$

Step 4: From v_3 move to adjacent vertex v_2 with higher objective: $z(v_2) = 2 - 0 = 2$

Step 5: Check adjacent vertices of v_2

Step 6: Terminate, declare the optimal vertex to be v_2 , and optimal value $z_{max} = 2$.

Summary

A **linear program (LP)** is an optimization problem that aims to maximize or minimize a linear objective function, subject to linear equality and/or inequality constraints.

Solving LPs:

Graphical method: works best when there are only two variables.

Simplex algorithm: efficient for large-scale LPs

Key insight: Optimal solutions occur on the boundary of the feasible region — typically at its vertices.

References

The graphical method:

<https://ncert.nic.in/textbook/pdf/lemh206.pdf>

The Simplex method:

<https://www.pnw.edu/wp-content/uploads/2020/03/attendance5-1.pdf>

Linear algebra basics:

<http://www.cs.cmu.edu/~zkolter/course/linalg>

Further readings

Linear programming tutorials:

<https://math.mit.edu/~goemans/18310S15/lpnotes310.pdf>

<https://www.uky.edu/~dsianita/300/online/LP.pdf>

<http://ndl.ethernet.edu.et/bitstream/123456789/20700/1/181.pdf>

